From Diagnosis to Solution

SLEEP: Implementation Milestone and Validation Plan

Interim progress report, ahead of empirical validation

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Professor Guha,

This interim report covers the three weeks since the proposal of 7 July. The engineering phase is complete: every code change the revision plan called for — the three-phase P1/P2/P3 roadmap and the retrieval-aware warm-up extension — is implemented and verified, with the full test suite green (**362 tests passing, 0 failing**, up from 297 before this phase). What remains is the GPU validation on RunPod, whose numbers will populate the final report. The report is deliberately honest about the boundary between the two: the code that *could* close the recognition–recall gap now exists and is correct; whether it *does* is the empirical question the runs will answer. Below: what was built, how it was verified, and the runs that will produce the results.

1. Progress against the plan

Every item from the proposal's roadmap is implemented and unit-tested on CPU (via tiny in-memory models), or, for the model/precision variants, staged as run-ready configuration. Nothing has yet been executed at 7B scale.

P1 — Critical	
Multi-seed harness (±SD aggregation) + seeding wired through the pipeline	**Done**
Extended multi-cycle (10–20 cycles via existing cycle/batch controls)	Ready
Multi-model & multi-precision configs (Llama 3, Mistral, Qwen 1.5B, float32)	Ready
P2 — High	
Standard-benchmark loaders (LAMA, CounterFact, real post-cutoff facts)	**Done**
Missing baselines: RAG, EWC-only, in-context injection	**Done**
Two-stage validation (surprise pre-filter + direct mini-recall gate)	**Done**
Biological-framing revision (constrain-or-trim)	Paper edit
P3 — Polish & Solution	
Retrieval-aware warm-up extension (gate + trainer + injector seam)	**Done**
Calibration plot (confidence vs. accuracy, tagged vs. untagged)	**Done**
Failure-mode table (C1–C5)	Paper edit

"Done" means implemented and covered by passing tests; "Ready" means the code path is exercised by tests and the run is a matter of configuration on the pod; "Paper edit" items are writing tasks folded into the final report. The two model/precision items are "Ready" rather than "Done" because the component paths are tested but the specific external models have not been loaded.

2. The extension is built — and verified where it can be

The centrepiece of the proposal was the retrieval-aware warm-up: a small trainable gate that teaches a frozen model to *use* key–value (KV) memory during generation, addressing the recognition–recall gap the paper diagnoses. It is now implemented as three pieces:

- A **memory gate** — a per-layer, per-head multiplicative scale on the memory value contribution (roughly 36 parameters for the top third of Qwen2.5-7B). Because attention is linear in the values, this exactly modulates how strongly retrieved memory feeds generation. It is initialised to identity, so an untrained gate changes nothing.
- A **warm-up trainer** that, on a general corpus (never the evaluation facts), stores a passage's prefix in memory and trains the gate to predict the continuation from it — the Memorizing-Transformers-style objective, teaching the *skill* of using memory, not any fact.
- An **injector seam** so the gate lives on the attention path rather than in the consolidation adapter, and therefore persists across sleep cycles instead of being overwritten by them.

Crucially, the change touches the KV-injection critical path, so it was verified against ground truth before anything was built on top of it. Per our standing practice, the *first* test written was a hook-based equality check: with an identity gate installed, the injected-attention output must be bit-identical to the ungated path. It is (to 10^{-6}). The 54 pre-existing KV-injection and memory tests continue to pass unchanged, and the gate is confirmed differentiable end-to-end so the warm-up can actually train it. In short, the mechanism is in place and provably does not perturb the verified recognition pathway until it is deliberately trained.

While there, we generalised the injector to resolve a model's attention module dynamically, so KV injection nominally runs on Llama and Mistral as well as Qwen. This is what makes the cross-model recognition test (P1) feasible; we flag in Section 5 that only the Qwen path is bit-verified so far.

3. Verification and rigor

The engineering is not asserted, it is tested. Concretely:

- **362 tests pass, 0 fail** — about 65 new tests over the prior 297, covering the gate, the warm-up loop end-to-end on a tiny in-memory model, the seeding utilities, the benchmark loaders, the two new baselines, the calibration plot, and the two-stage-validation gate.
- **Ground-truth-first on the critical path.** The identity-gate equality check and a gate-gradient check were written before the trainer, so a silent change to injection could not masquerade as a warm-up effect.
- **The self-grading gap is now instrumented.** Two-stage validation records how many candidates pass the surprise proxy versus the direct free-form recall gate, so the C4 discrepancy the paper names is measured as a number rather than argued.
- **Reproducibility built in.** A single call seeds every RNG the pipeline touches, and the multi-seed runner executes an experiment across seeds in independent processes and reports mean ± standard deviation — directly answering the review's "no single-seed results" requirement.

Everything above runs on CPU, so correctness was established without spending GPU budget. The 7B runs are for measurement, not debugging.

4. What the validation runs will decide

The implementation cannot, by itself, tell us whether the extension works — only whether it is correct. The scientific claims remain open until the runs land, and the report is explicit about that. The pre-registered success criterion is unchanged from the proposal: if warmed-up SLEEP reaches DRA $\approx$ 0.10–0.15 at BCP < 1.5 on the 200-fact benchmark, the recognition–recall gap is closable within the framework. The runs, in priority order and with the number each produces:

1. **Multi-seed recognition & Pareto** (seeds 0–4): does the +0.16 recognition signal and the single-cycle frontier survive seed variance, as mean ± SD?
2. **Extended multi-cycle** (10–20 cycles, seeds 0–4): does base- capability preservation plateau or keep climbing beyond three cycles?
3. **Warm-up extension** — the headline: the SLEEP-with-warm-up row of the new results table (DRA, BCP) against the without-warm-up baseline.
4. **float32 and a second model family**: is the C1 precision-floor finding a bfloat16 artefact, and does the gap generalise beyond Qwen?
5. **Standard benchmark & baselines**: the three baselines and an established dataset, for

comparability with prior work.

Estimated compute is roughly USD $8–15 depending on the seed and cycle counts, within the pod budget. Each run writes a machine-readable result file, so the final report is a matter of tabulating outputs, not re-deriving them.

5. Open risks

Three, stated plainly:

- **The extension may only narrow, not close, the gap.** A partial rise in DRA is still a publishable diagnostic-to-treatment result, but the magnitude is unknown until the run. We are not assuming the modest success criterion will be met.
- **Cross-model KV injection is CPU-verified for Qwen only.** The Llama/Mistral attention path is generalised but not yet bit-checked on hardware; the first cross-model run doubles as that check, and the lower-risk float32 and multi-cycle comparisons do not depend on it.
- **Compute state.** The pod's connection details and remaining credit change between sessions; we will confirm both before starting, and lead with the cheapest high-signal run (the warm-up on a 50-fact subset, about $1) as a smoke test before committing to the full seed sweeps.

6. Revised timeline

Stage	Deliverable	Status
1. Proposal	Roadmap + extension, sign-off request	**Done** (7 Jul)
2. Implementation	All code + tests; this report	**Done** (27 Jul)
3. Validation	GPU runs on RunPod (§4)	Pending GPU (∼1 week)
4. Final report	Results, tables, paper revision	After runs

The project is on the plan submitted three weeks ago. The build phase came in where scoped; the only work between here and the final report is executing the runs and writing up what they show.

7. Requested input

No decision is required to proceed — the runs are queued and can begin as soon as the pod is available. Two light confirmations would help us aim the final report:

1. That the validation priority order in Section 4 matches how you would weight the evidence, or your steer if not.
2. A target venue and deadline, so the final report is written to the right format and length from the start.

We will report the smoke-test result as soon as it is in, and the full tables in the final report to follow.

Respectfully submitted — Aditya Tripathi. Implementation, tests, and experiment scripts are in the project repository; the 7 July proposal accompanies this report.